



# AgentOps in production

Agentic end-to-end observability with Red Hat AI

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# The evolution of enterprise AI

How it started

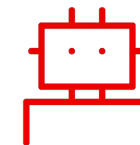
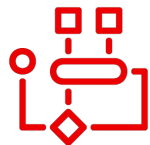
Where it is!

Where it is going...

LLM **Workflow**

Agentic **Workflow**

Autonomous **Agents**



The LLM is a simple worker,  
and autonomy is **Low**.  
Orchestration is external.

The LLM has decision  
making power, and autonomy  
is **Medium**.

Given a high-level objective,  
the agent defines the tasks  
and plans. Autonomy is **High**.

# Agentic AI is no longer experimental

it's operational, scaling fast, and redefining how work gets done.

**80%**

**Fortune 500  
adoption**

of Fortune 500  
companies already  
use active AI agents.

– Microsoft Cyber  
Pulse, 2026

**\$632B**

**AI spending  
by 2028**

Worldwide spending  
on AI solutions will  
surpass \$632 billion  
by 2028, growing at  
29% CAGR.

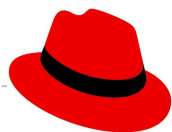
– IDC, 2025

**82%**

**Executive intent**

Tech executives plan  
to integrate AI  
agents within 1–3  
years, but <10% have  
production-grade  
deployments.

– Capgemini,  
2025



# Red Hat AI



## Inference

Increase efficiency with **fast, flexible and efficient** inferencing



## Data

Simplified experience for **connecting data to models and agents**



## Agents

**Scale Agentic AI** with governance and control

## Platform

**Governance and security from metal to agents**

Run any model or agent across any hardware in any cloud environment

# Three critical gaps: Pilot to production

## Agent identity

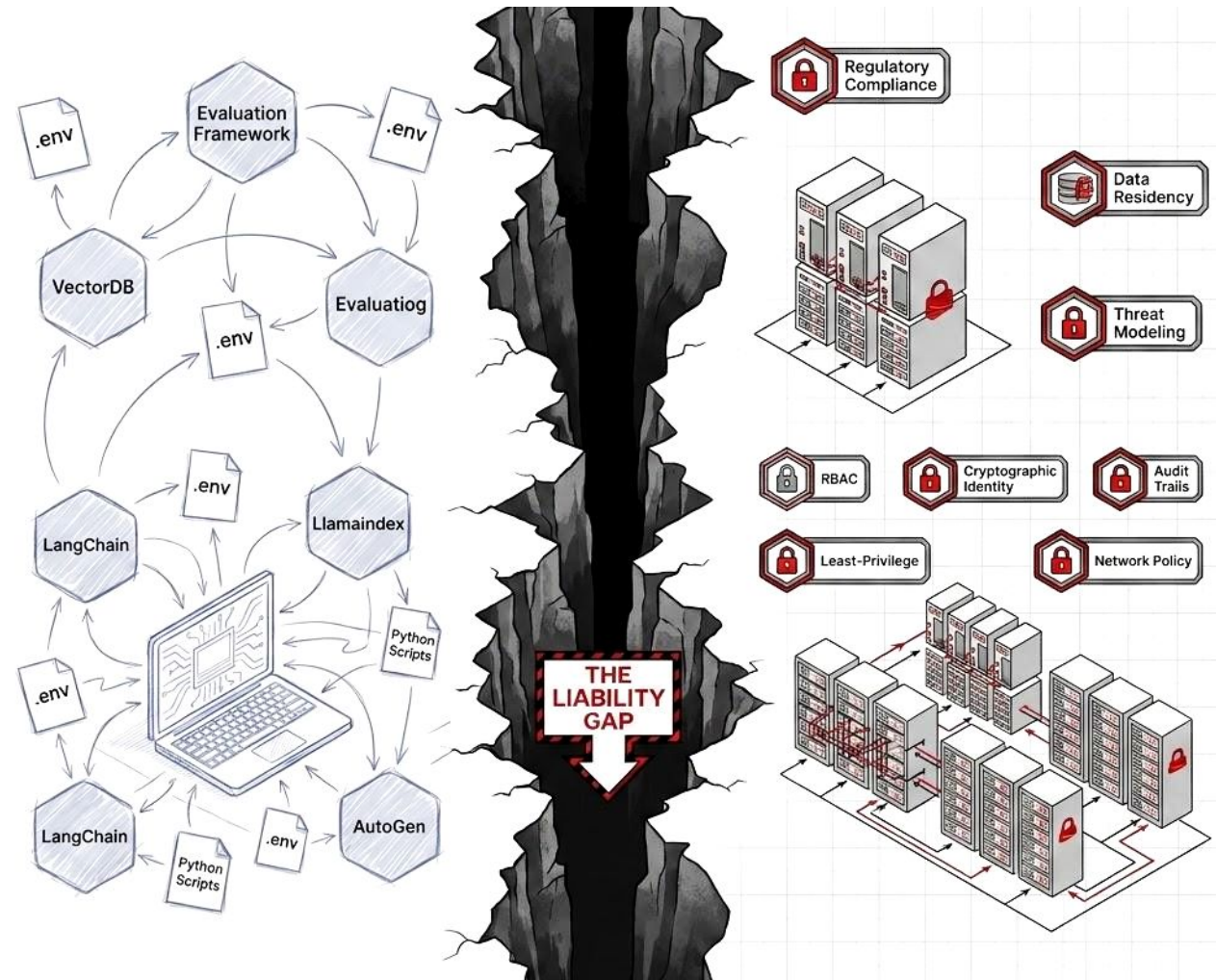
Agents need secure, cryptographic identities to access sensitive tools under least-privilege principles – hardcoded keys are unacceptable in production

## Ungoverned autonomy

Unconstrained dynamic code execution creates security risks, requiring isolated environments with deep execution tracing to monitor autonomous actions

## Scalability and performance

Multi-agent architectures generate unpredictable, concurrent inference spikes that traditional infrastructure cannot handle



# Streamline Agent Operationalization

## Security



### Agent Identity & Security

Govern what agents can access

- ❑ **Cryptographic Identity:** Short-lived, keyless SPIFFE/SPIRE
- ❑ **Infrastructure Defense:** MCP Gateway blocking
- ❑ **Scoped Access:** Managed OAuth2 tokens.

## Safety



### Agent Sandbox

Isolate and contain agent execution

- ❑ **Zero-Trust Sandboxing:** Kernel-isolated microVMs.
- ❑ **Egress Control:** Unauthorized data prevention
- ❑ **Scalable Inference:** High-throughput vLLM scaling.
- ❑ **Binary Isolation:** OpenShell network/syscall filtering.

## Observability



### AgentOps

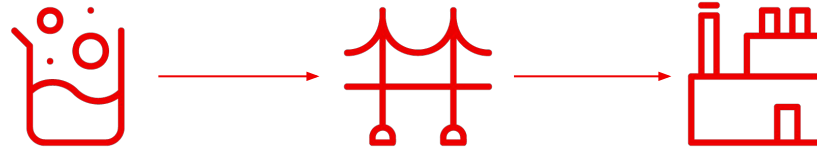
Monitor, trace, and manage the lifecycle

- ❑ **MLflow tracing:** logs every LLM call, tool execution, and decision step (GA)
- ❑ **Agent Lifecycle:** Kagenti discovery/scaling.
- ❑ **Live Scoring:** EvalHub continuous evaluation.

# AgentOps with Red Hat AI

Bridge the gap between experimental AI to Agents in production

**What is it:** A unified approach for managing the full lifecycle, security, and performance of AI agents and their tools



**Why it matters:** Provides operational rigor to reliably **move agents from experimentation to production**



Tracing



Evaluation



Identity



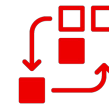
Safety/Security



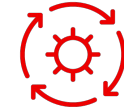
Observability



Catalog



Registry



Lifecycle Mgt

## Observe, Trace, Evaluate

- **MLFlow 3.10+**
- Log **traces** and **visualize** them including tool calls (3.4)
- **OpenTelemetry** compatibility
- Datasets curation + Scoring with **Eval-hub** and **MLFlow**

## Enforce Identity/Access Control

- **OAuth2 / Token Exchange** for secure delegation
- **SPIFFE/SPIRE** for trusted workload (agent) identity
- **Kagenti/AuthBridge** for agentic Auth and access control
- Agent to tools governance (MCP Gateway)

## Develop and Deploy Reliably

- The same **Cloud-native** principles **repurposed** for agentic workloads
- **Manage Agent Lifecycle with declarative**, platform-managed Custom Resources via Kagenti (runtime, identity, trace config, etc).
- **Agent safety**, guardrails, and red teaming (Nemo, Garak, etc)

# AgentOps Lab







**Red Hat AI**



# Thank you



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